# ASHVATH V CHEPPALLI

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## EDUCATION

**Northeastern University,** Boston, MA — Sep. 2025 - May 2027 (Expected)
*Master of Science in Artificial Intelligence* — GPA: 3.83
- *Relevant Coursework:* Foundations of AI, Algorithms, Machine Learning, Neural Networks, NLP, Computer Vision

**Rajalakshmi Engineering College,** Chennai, India — 2021 - 2025
*Bachelor of Technology in AI and Machine Learning*
- *Relevant Coursework:* Deep Learning, Computer Vision, Reinforcement Learning, Data Structures & Algorithms

## PROJECTS

**Layer-Aware Quantization Pipeline for Efficient DistilBERT Deployment** — Project Link
- Built **INT8-PTQ** pipeline for **DistilBERT** with **2.19x** CPU speedup and **64%** less memory at only **0.5%** accuracy drop
- Profiled INT8 sensitivity of **38** Linear modules one at a time and found the 3rd FFN block the most fragile
- Benchmarked **4** precision modes and found INT8 stayed within **1%** of FP32 under **3** noise types
- Shipped a **Streamlit** dashboard with an INT8 cost savings calculator backed by **24 pytest** tests

**Intelligent Human Detection for Emergency Fire Evacuation** — Project Link
- Fine-tuned **YOLOv8** for human detection in thermal imagery with **89.6% mAP50** and **86.7% precision**
- Built a **Flutter** mobile app running **YOLOv8** at **24 FPS** for real-time human detection by rescue teams
- Hand-annotated **4,000+** thermal frames across **12** fire scenarios to build the training dataset

**AI-Optimized Energy Consumption Forecasting System** — Project Link
- Trained an **XGBoost** regression model on **1M+** records with **0.038 kWh MAE** and **95%** lower error than baseline
- Performed EDA and feature engineering in **Pandas** to identify peak consumption patterns for optimization
- Served the forecasting model through a **REST API** endpoint with under **120ms** prediction latency

## WORK EXPERIENCE

**Skillible,** *AI Intern*, Chennai, India — Jul. 2024 - Aug. 2024
- Benchmarked **6** bias mitigation methods for generative AI models including re-weighting and adversarial debiasing
- Evaluated **transformer** model performance across NLP tasks using **Python** and **Scikit-learn** to guide model selection
- Designed **15+** prompt templates to improve **LLM** output quality across internal use cases

**Anugraha Systems (Rajalakshmi Engineering College),** *Web Development Intern*, Chennai, India — Jan. 2023
- Built full-stack web apps in **React** and **MySQL** and redesigned frontend to backend data flow for **4** workflows
- Shipped a customer-facing **dashboard** with real-time data display
- Delivered end to end applications across the full software development lifecycle from requirements to deployment

## TECHNICAL SKILLS

- **Computer Languages:** Python, SQL, JavaScript
- **ML/AI:** PyTorch, TensorFlow, Hugging Face, Scikit-learn, OpenCV, YOLOv8, XGBoost, LangChain
- **Data:** Pandas, NumPy, Matplotlib
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Cloud & DevOps:** AWS (EC2, S3), Docker, Git, GitHub
- **Frameworks & Tools:** FastAPI, Flask, React, REST APIs